Approaches and Techniques for Extracorporeal Membrane Oxygenation

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EXTRACORPOREAL MEMBRANE OXYGENATION BASICS

- Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support is a modality that allows for the partial or complete replacement of the cardiopulmonary system. The patient's venous blood is drained and passed through a centrifugal pump and then through an oxygenator that facilitates the oxygenation of blood and the removal of carbon dioxide. This blood is returned to the patient either via the arterial system in the case of full cardiopulmonary support or to the venous system in the case of isolated pulmonary support.

Veno-arterial Extracorporeal Membrane Oxygenation

- Veno-arterial ECMO (VA-ECMO) refers to full cardiopulmonary support.[1] The most common indications for VA-ECMO include postcardiotomy shock following cardiac surgery, cardiogenic shock, cardiac arrest requiring cardiopulmonary resuscitation, massive pulmonary embolism with cardiorespiratory collapse, acute poisonings, and hypothermia. It is contraindicated in the context of an irreversible disease process, poor preexisting functional status, multisystem organ failure, and severe aortic valve insufficiency.
- Cannulation for VA-ECMO is accomplished either centrally via the ascending aorta or axillary artery (reinfusion) and right atrium (venous drainage), usually via a median sternotomy, or peripherally, typically via the femoral vessels (Fig. 32.1). The choice of cannulation sites will depend on the patient's anatomy, clinical status, vascular access, and hemodynamic needs.

Veno-arterial Extracorporeal Membrane Oxygenation Pearls and Pitfalls

Differential Oxygenation

- Differential oxygenation, often also referred to as Harlequin syndrome or North-South syndrome, is a phenomenon that occurs in the setting of peripheral cannulation when there is recovery of cardiac function but concomitant respiratory failure. Cerebral and coronary oxygenation depend on the location of the arterial reinfusion cannula and in the setting of peripheral cannulation, retrograde blood flow provides oxygenated blood to the upper body. As native contractility improves, the mixing point of native circulation and retrograde reinfusion circulation occurs more and more distally along the aorta. In the setting of ongoing respiratory failure, the blood ejected from the heart into the coronary bed and head vessels is oxygen-poor blood that has bypassed the extracorporeal circuit. Patients with peripheral VA-ECMO cannulation must be monitored carefully for this phenomenon via monitoring of the arterial saturation from the right radial or brachial artery, as well as continuous cerebral oximetry if available. If differential oxygenation is detected, a trial of higher flows can be performed to determine if oxygenation improves; otherwise, most often an additional cannula into the superior vena cava (SVC) must be added, usually via cannulation of the internal jugular vein resulting in a veno-arterial-venous or V-AV configuration that allows delivery of oxygenated blood into the venous system (Fig. 32.2).

Left Ventricular Distention

- While on VA-ECMO, efforts should be made to ensure adequate left ventricular (LV) contractility to allow for ventricular unloading.
- Drainage from the thebesian and bronchial veins into the LV can accumulate and cause distention in a nonejecting LV, resulting in an elevated left ventricular end diastolic pressure, which in turn will result in decreased coronary perfusion and subendocardial myocardial ischemia, further hindering LV recovery.
- Blood stasis can result in the development of LV or aortic root thrombus. Risk factors for LV distention on VA-ECMO include aortic insufficiency, inadequate systemic venous drainage (which increases pulmonary venous return), arrhythmias, and systemic arterial hypertension. LV distention will commonly present as refractory ventricular arrhythmia, pulmonary edema or hemorrhage, refractory hypoxemia, or simply an elevated pulmonary capillary wedge pressure. Pulmonary artery diastolic pressure greater than 8 to 10 mm Hg while on full-flow VA-ECMO is usually suggestive of inadequate LV unloading. Echocardiographic examination may demonstrate the absence of aortic valve opening or evidence of blood stasis. Maintenance of an arterial pulse pressure of >10 mm Hg is generally thought to represent adequate pulsatility to allow for LV ejection. This can be achieved with inotropic therapy, afterload reduction, ensuring AV synchrony, and an increase of positive-end expiratory pressure on the ventilator.
- If these conservative measures are insufficient to ensure adequate pulsatility, consideration should be given to mechanical LV unloading with aortic counterpulsation, temporary transvalvular microaxial flow pump, atrial septostomy, or surgical placement of a cannula in one of the pulmonary arteries, right superior pulmonary vein, or LV apex to unload the left ventricle. Ideally, LV unloading or venting should be accomplished within 12 hours to optimize outcomes.[2]

Veno-venous Extracorporeal Membrane Oxygenation

- Veno-venous ECMO (VV-ECMO) refers to isolated pulmonary support and is most commonly utilized in cases of respiratory failure secondary to acute respiratory distress syndrome, primary graft dysfunction after lung transplantation, acute exacerbation of lung disease, and traumatic lung contusion. It is contraindicated in the setting of irreversible neurologic, irreversible lung injury if not a candidate for lung transplantation, severe pulmonary hypertension, life expectancy <6 months, cardiac arrest, advanced liver disease,

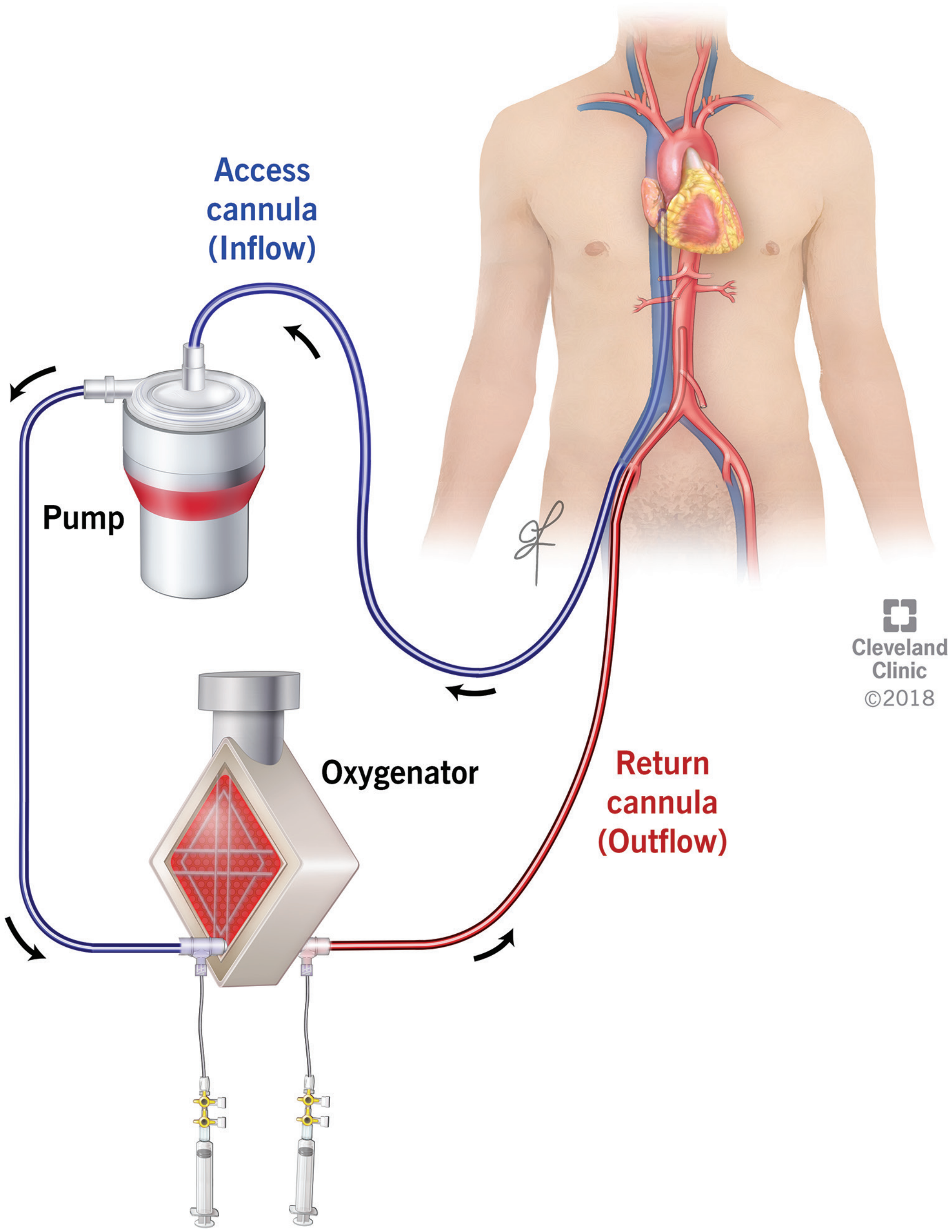

Fig. 32.1

advanced multiorgan failure, and in cases of inadequate vascular access.

- In the setting of respiratory failure, VV-ECMO is indicated in cases of refractory hypoxemia or hypercarbia despite optimized mechanical ventilation, and usually after a failed trial of prone positioning. VV-ECMO improves oxygenation and ventilation, all while reducing ventilator-associated lung injury and enabling lung recovery. Widely accepted thresholds for VV-ECMO therapy, as published by ELSO, are as follows:[3]
  1. Hypoxemic respiratory failure with P/F ratio <80 mm Hg after medical management, including a trial of prone positioning
  2. Hypercapneic respiratory failure with pH <7.25 despite optimal conventional mechanical ventilation (respiratory rate > 35 and plateau pressure ≤30 cm H_20)
  3. Ventilatory support as a bridge to lung transplantation or primary graft dysfunction following lung transplant
- Two common configurations of VV-ECMO cannulation are most routinely utilized—a dual-site, two-cannula configuration that drains blood from the inferior vena cava (IVC) with reinfusion via the right atrium (RA) (femoral vein-femoral vein or femoral vein-internal jugular vein cannulation) (Figs. 32.3 and 32.4) or a single-site, dual-lumen cannula inserted via the internal jugular or

Fig. 32.1 Peripheral cannulation veno-arterial extracorporeal membrane oxygenation. (Reprinted with permission, Cleveland Clinic Foundation ©2024. All Rights Reserved.)

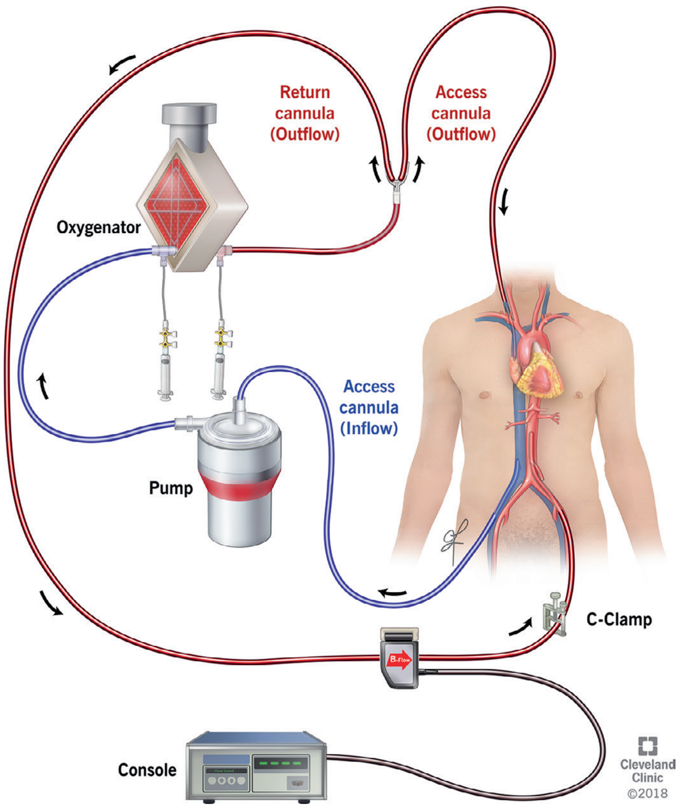

Fig. 32.2

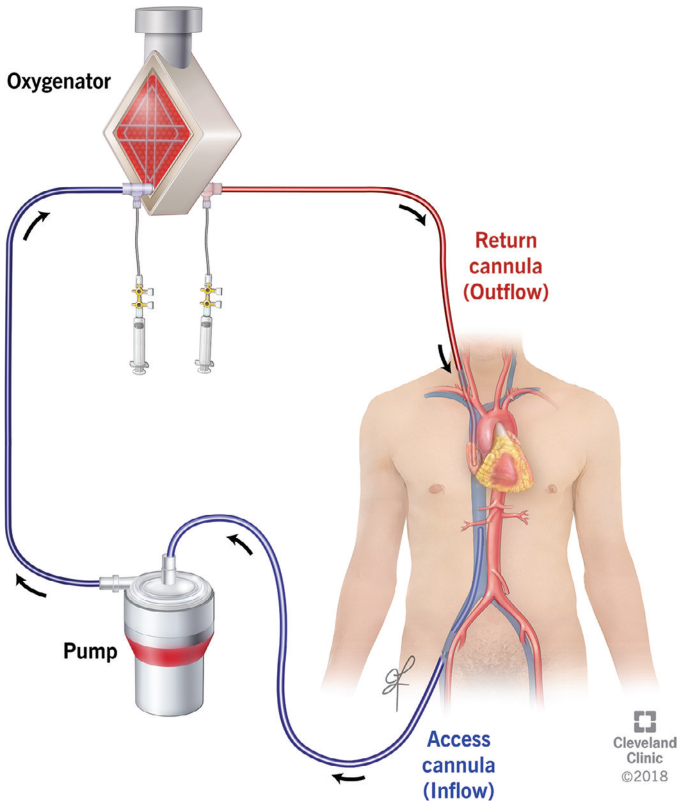

Fig. 32.4

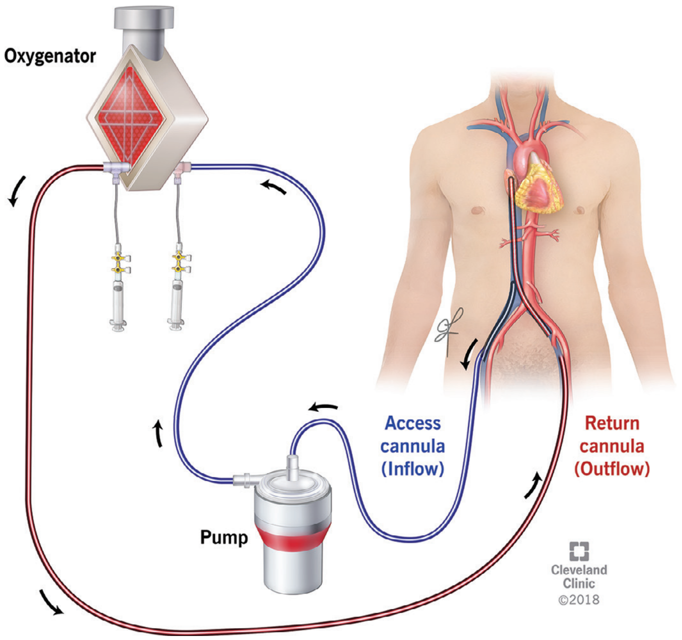

Fig. 32.3

subclavian vein with simultaneous IVC and SVC drainage via one lumen and reinfusion into the RA via a second lumen (Fig. 32.5).

- Single-site cannulation is commonly achieved with an Avalon or Crescent catheter in the right internal jugular vein or subclavian vein. The use of a dual-lumen catheter reduces the risk of recirculation, infection, and cannula dislodgement. It has the advantage of allowing for patient mobilization, which can allow for expedited recovery in the critically ill patient. The disadvantage of single-site cannulation is the limitation in individual inner cannulae size, which can limit the amount of flow and subsequently, the amount of support that can be provided. The overall larger diameter of the cannula can predispose to a higher risk of thrombotic complications.

Veno-venous Extracorporeal Membrane Oxygenation Pearls and Pitfalls

Recirculation

- Dual-site cannulation carries the risk of recirculation—a phenomenon in which oxygenated blood is drawn by the drainage cannula. This results in a lack of oxygenation to the right ventricle and systemic circulation and it also reduces the efficiency of the circuit. This occurs more commonly with a femoral vein-internal jugular configuration and can usually be mitigated by ensuring a distance of at least 10 cm between the two cannulas during placement, with routine checks of positioning during the patient's course with bedside radiographs. Reductions in flow, if tolerated, can also minimize recirculation. In the setting of femoral vein-femoral vein, this is of less concern as a short drainage cannula is placed in one vein while a long reinfusion cannula is placed in the contralateral femoral vein.

Inadequate Flow

- Occasionally, higher flows are required and a two-cannula system is inadequate to maintain the level of gas exchange that is required. This is the case in the setting of very high cardiac output and oxygen extraction or a patient with a higher body mass index.

Fig. 32.2 Veno-arterial-venous extracorporeal membrane oxygenation configuration. (Reprinted with permission, Cleveland Clinic Foundation ©2024. All Rights Reserved.)

Fig. 32.3 Veno-venous extracorporeal membrane oxygenation with bifemoral venous cannulation. (Reprinted with permission, Cleveland Clinic Foundation ©2024. All Rights Reserved.)

Fig. 32.4 Veno-venous extracorporeal membrane oxygenation with internal jugular and femoral vein cannulation. (Reprinted with permission, Cleveland Clinic Foundation ©2024. All Rights Reserved.)

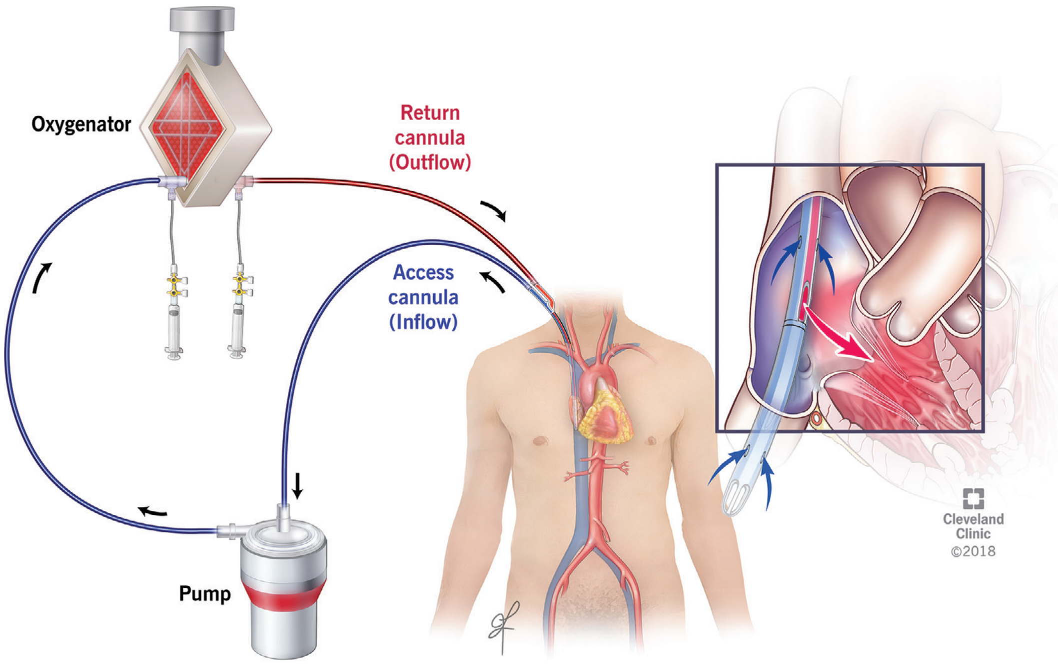

Fig. 32.5

If conservative measures such as beta-blockade and sedation are inadequate to reduce cardiac output, an additional drainage cannula may be added. This typically necessitates an internal jugular reinfusion and bilateral femoral drainage cannula configuration.

OPERATIVE TECHNIQUES

General Procedural Considerations

Cannula Selection

- Various cannulas are available on the market from various manufacturers. Availability and selection will be determined by local institutional availability, but they will generally function similarly with only nuanced differences in performance. Over- or undersizing can result in venous obstruction, injury to the vessels, or inadequate flow. It is advisable to consult manufacturer-provided flow tables as well as to engage with your perfusion team to determine the optimal cannula size and type for your patient with consideration to vessel size in the patient, if known. In general, short cannulas with larger internal diameters or long cannulas with multiple drainage holes will allow for higher flows. In the average-sized adult patient, a reinfusion cannula sized 18- to 20-F and a drainage cannula size 24- to 28-F are usually sufficient to provide drainage and flow up to 5 L.

Circuit Management

- After cannula placement, clamps are applied. Circuit tubing should be brought up to the field in a sterile fashion and prepared for connection. It is imperative to ensure for adequate length of tubing to ensure that the lines are left free of tension. Connection of the circuit tubing to the cannulas should ensure complete de-airing of the connections with the use of sterile saline. Once the clamps are removed, the tubing should be inspected to ensure complete de-airing. Following this, circuit flow can commence. Following the commencement of flow, a safety pause should take place to ensure correct cannula configuration and adequate color differential between patient inflow and outflow cannulas.

Veno-arterial Extracorporeal Membrane Oxygenation Cannulation

Central Aortic Arterial and Right Atrial Venous Cannulation

- Since central cannulation of the ascending aorta and right atrium for ECMO follows the same general principles for cannulation for cardiopulmonary bypass, it will not be covered in detail here. For a detailed discussion of techniques for central cannulation, please refer to Chapter 2. In brief, an appropriate site for aortic cannulation, free of calcification, is selected. Two diamond purse strings are created using 4-0 prolene sutures and engaged with tourniquets. These sutures can be pledgeted at the discretion of the surgeon if there is a concern for friable aortic tissue or suspected challenging hemostasis. An No. 11 or 15 blade is used to make an aortotomy large enough to accommodate an appropriately sized cannula, and the cannula advanced slowly and gently into the ascending aorta, and secured with the tourniquets. Venous cannulation is achieved by placing a 4-0 prolene purse string either on the face of the right atrial appendage, or around the right atrial appendage. The atrium is incised with an No. 11 blade, the trabeculae cleared with Metzenbaum scissors or spreading of a blunt end tonsil, and the cannula placed into the right atrium and the tip positioned in the IVC. It is important to recognize that a generous atrial cuff around the right atrial venous cannula must be made to ensure no entrainment of air.
- Alternatively, a configuration with central aortic cannulation, and femoral venous cannulation can be undertaken. Femoral venous cannulation is discussed in detail in the sections to follow.
- Central cannulation allows facile insertion of a surgical vent if LV venting is needed. A 16-F or 20-F cannula can be inserted via the right superior pulmonary vein into the left ventricle and secured with a purse string. The LV apex can also be cannulated with a similar-sized cannula. Finally, a vent in the pulmonary artery can be useful in the case of inadequate right-sided drainage. The LV vent should be Y-ed off to the venous return tubing.
- In many cases, the chest is left open when a patient is centrally cannulated, but in some cases, it may be appropriate to position the

Fig. 32.5 Single-site veno-venous extracorporeal membrane oxygenation cannulation via right internal jugular vein. (Reprinted with permission, Cleveland Clinic Foundation ©2024. All Rights Reserved.)

central cannulas through a counterincision subcostally and approximate the sternum. This comes with a theoretical reduction in the risk of infection and the potential ability to ambulate an awake patient on VA-ECMO.[4] This approach is a consideration if longer-term support is anticipated, such as in the scenario of ECMO as a bridge to transplant. Technically, this is achieved by creating counter incisions subcostally and creating tunnels through which the distal ends of the cannulas can be pulled through, and the circuit tubing then connected.

Axillary Artery Cannulation

- A transverse incision is made roughly 2 cm below the clavicle in the deltopectoral groove. Self-retaining retractors are usually required to optimize exposure. The pectoralis major muscle is dissected down to the levels of the clavipectoral fascia, which is then incised to expose the pectoralis major muscle. This is either retracted laterally or dissected.
- Below the pectoralis major should lie the axillary artery and vein (Fig. 32.6). The artery is superior and deeper to the vein. If necessary for exposure of the artery, the vein can be encircled with a vessel loop and retracted toward the head. The artery should be dissected out and encircled proximally and distally with vessel loops. Note that the medial and lateral cords of the brachial plexus can be nearby, or at times overlying the artery, and care must be taken to avoid injury to them.
- Heparin should be administered to achieve an ACT >200 seconds. The artery should be controlled proximally and distally with clamps and an arteriotomy made to accommodate an 8-mm tube graft, which is then anastomosed to the artery in standard fashion using a 5-0 prolene suture. There are then two options:
- One approach would see an appropriately sized arterial cannula placed inside the tube graft to the level just beyond the anastomosis but not advanced fully into the axillary artery itself. The cannula is secured in the tube graft with silk ties tied at three to four points along the length of the cannula and tube graft (Fig. 32.7). The cannula is then connected to the arterial limb of the circuit tubing.
- Another approach would see a connector fastened to the end of the tube graft using multiple silk ties, and the circuit connected directly to the connector. At the time of decannulation, the graft can be controlled just above the anastomosis with large hemoclips, trimmed, and the stump then oversewn with a 5-0 prolene suture. Although the axillary artery is an attractive site for peripheral cannulation, there is a risk of upper ipsilateral limb hyperperfusion, which must be mitigated either with placement of a cannula inside the graft and slightly past the anastomosis to limit antegrade flow or to ensnare the distal axillary artery.

Femoral Artery Cannulation via Cut Down

- After identification of the femoral artery by palpation of the pulse, either a vertical incision is made over the top or an oblique incision is made aligned with the inguinal ligament.
- Dissection should be carried down to the femoral artery and vein with careful attention to identify its branches and the superficial

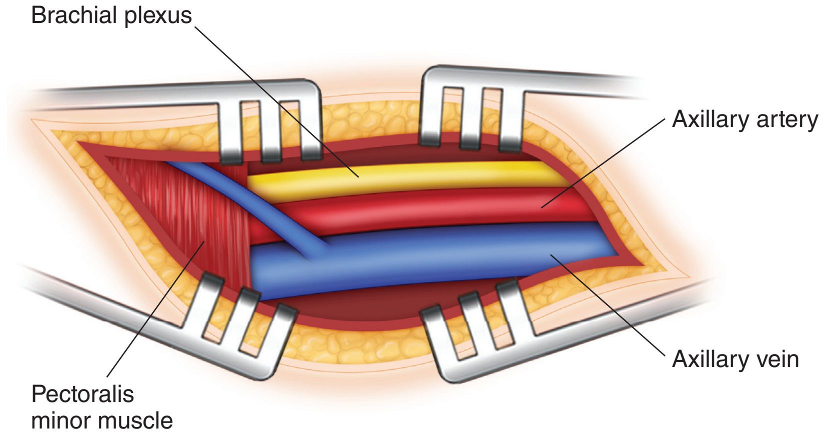

Fig. 32.6

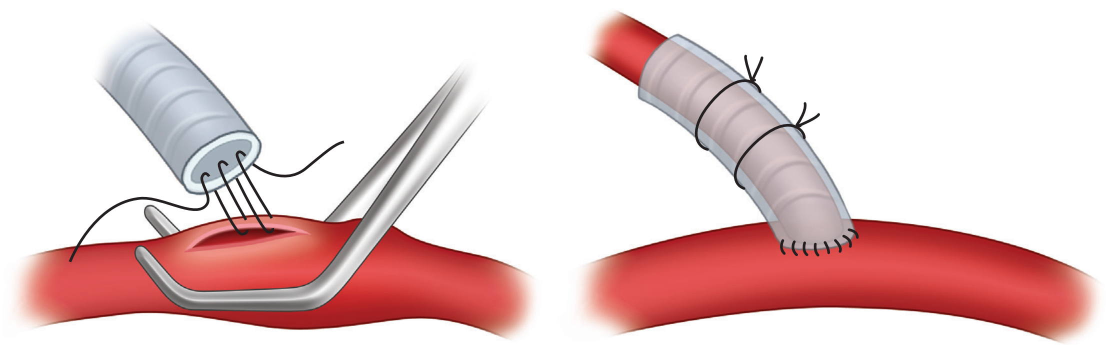

Fig. 32.7

Fig. 32.6 Axillary artery exposure.

Fig. 32.7 Anastomosis of tube graft to axillary artery for axillary artery cannulation.

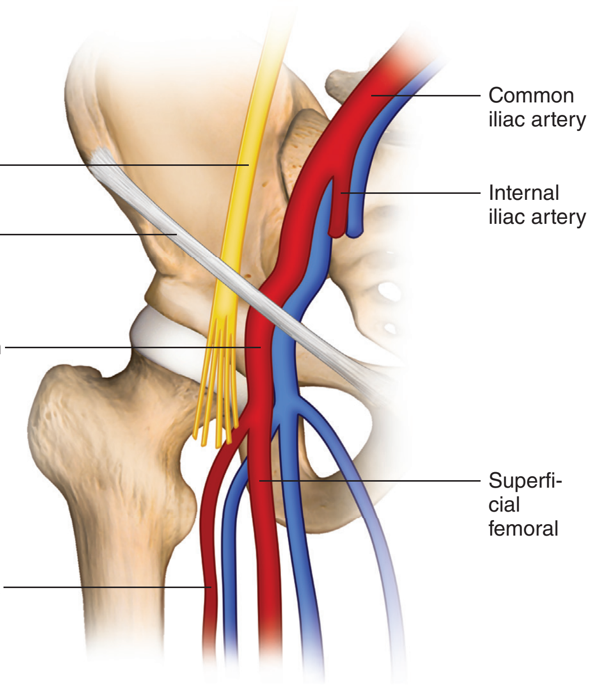

Fig. 32.8

femoral artery and profunda femoris to ensure cannulation of the common femoral artery (Fig. 32.8).

- Proximal and distal control are obtained with the use of vessel loops. The site selected for cannulation and clamping should be free of any significant calcification. Heparin is be administered to target ACT >200 seconds. The proximal artery should be clamped, and the distal artery should be either clamped or occluded with a vessel loop. A purse string with a 5-0 prolene is created and secured onto a tourniquet on the anterior aspect of the artery.
- Seldinger technique is used to advance a cannula over a guidewire after progressive sequential dilation into the common femoral artery. The cannula is then secured by tightening the purse string tourniquet (Fig. 32.9). The cannula is connected to the circuit tubing. The cannula and its tubing should then be secured at the insertion point and to the thigh at multiple points to prevent undesirable movement or dislodgement of the cannula. During decannulation, the femoral artery should be repaired primarily.

Femoral Venous Cannulation via Cut Down

- The femoral vein is identified medial to the femoral artery, and similarly, a purse string of 5-0 prolene is placed. Femoral cannulas are usually multistage for best drainage and are typically placed over a wire under either fluoroscopic or TEE guidance.
- A needle is inserted into the center of the purse string, followed by the insertion of the guidewire, which is advanced into the SVC.
- The cannula is then placed after either a venotomy large enough to pass the cannula or after progressive sequential dilation. The cannula is then advanced under imaging guidance to ensure placement with the tip just inside the SVC at the SVC-RA junction. Similarly, the cannula is connected to the circuit tubing, and the cannula and the circuit tubing are anchored to the insertion site and thigh at multiple locations.
- During decannulation, the venous purse string can be tied.

Percutaneous Femoral Cannulation

- Percutaneous femoral cannulation is often pursued in an emergency setting. Ideally, access is obtained under ultrasound guidance to ensure proper cannula placement.[5]
- It is advisable to gain wire access in the venous system first, then the arterial system. If a failed attempt is made at arterial access first, hematoma from an arterial puncture in the subcutaneous tissues can make ultrasound imaging of the vessels challenging for subsequent attempts.
- Peripheral venous and arterial cannulation is undertaken with the Seldinger technique. Prepackaged kits with puncture needles, wires, and dilators are readily available and convenient for use.
- After wire access is obtained, the vessels are sequentially dilated to a size adequate for the cannula size selected. Venous cannulation is ideally fluoroscopically or TEE guided, but in the case of emergency cannulation in a cardiac arrest scenario where image guidance is not feasible, placement can be visually estimated with the cannula against the patient with attention to advancing to a distance that correlates with this predetermined measure.

Distal Limb Reperfusion Cannulation

- In cases of peripheral femoral cannulation, the arterial cannula can obstruct the lumen of the common femoral artery sufficiently to prevent blood flow to the lower extremity. It is, therefore, advisable to place a catheter antegrade into the superficial femoral artery with ultrasound guidance and the Seldinger technique. A catheter between 6-F and 9-F is usually sufficient, with consideration of patient and vessel size. This catheter is connected to the arterial limb of the circuit.

Veno-venous Extracorporeal Membrane Oxygenation Cannulation

Femoral Vein Cannulation

- Femoral cannulation for VV-ECMO proceeds as described above. In the case of femoral-femoral VV-ECMO, both right and left femoral veins are cannulated.

Percutaneous Internal Jugular Vein Cannulation

- When cannulating for a femoral-internal jugular configuration, cannulation of the IJ is performed by ultrasound-guided puncture of the right internal jugular vein and passage of a stiff J-tip wire into the IVC under TEE or fluoroscopic guidance. The vessel is sequentially dilated and the cannula is advanced. A cannula with side holes within a short segment at the end of the cannula should be positioned with the tip at the SVC-right atrial junction. A cannula with a long segment of side holes at its tip should be positioned into the IVC.
- If the IJ is being cannulated for placement of a dual-lumen cannula, positioning is especially critical. The Avalon and Crescent cannulas have three ports. The most distal port at the tip of the cannula is meant to lie below the IVC-right atrial junction to facilitate drainage of the lower body. The most distal port is positioned in the SVC to facilitate upper-body drainage. The middle port serves as the reinfusion port and should be positioned in the right atrium and directed toward the tricuspid valve (Fig. 32.5).

Percutaneous Subclavian Vein Cannulation. Cannulation of the subclavian vein is an additional option for reinfusion for VV-ECMO, and cannulation of the left subclavian vein specifically is a desirable site for cannulation with a dual-lumen cannula, as it allows for patient comfort and neck mobilization. It is undertaken with traditional anatomic landmarks (Fig. 32.10). The sternal notch and angle of the clavicle are palpated, the latter usually roughly two-thirds of the length of the clavicle. The puncture needle with the syringe attached is inserted just below the angle of the clavicle and advanced, angled toward the sternal notch, guiding the needle below the clavicle, all while gentle

Fig. 32.8 Femoral vessel anatomy.

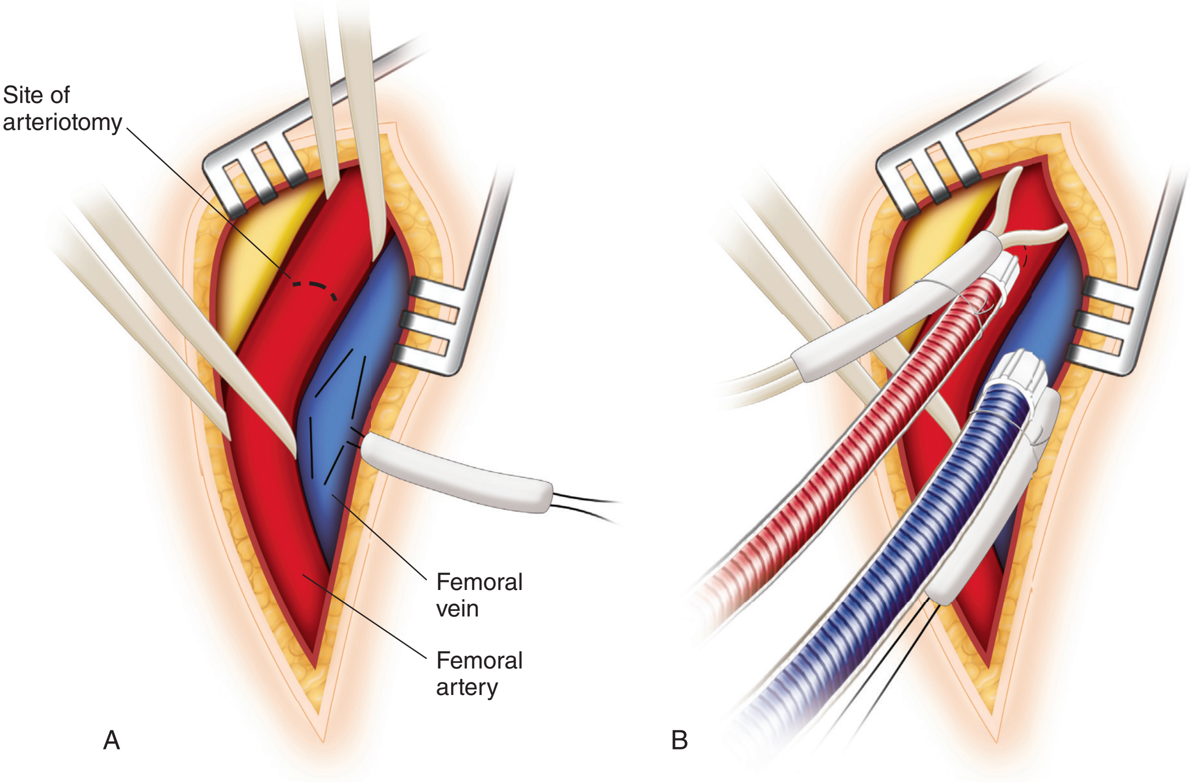

Fig. 32.9

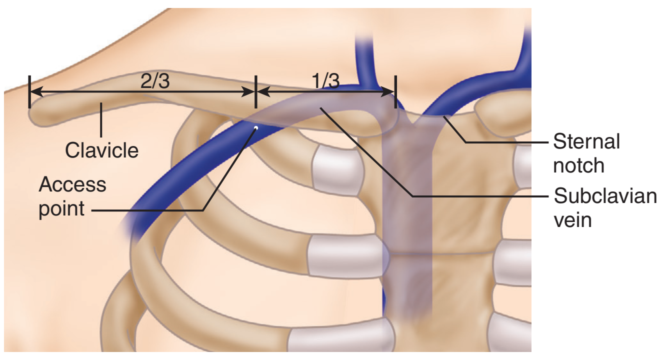

Fig. 32.10

suction is applied to the syringe. After a successful vessel puncture, a super-stiff wire is advanced under imaging guidance into the IVC. The vessel is sequentially dilated up to the desired cannula size and the cannula is placed and positioned as above.

Postoperative Care and Daily Best Practices

While discussion regarding postoperative care of the ECMO patient is beyond the scope of this chapter, it is important to highlight that ECMO carries the risk of significant morbidity, including the risk of bleeding, thromboembolic events, infection, and limb ischemia. It is imperative, therefore, that providers assess organ recovery regularly and wean ECMO support as soon as the patient tolerates it. Institutional protocols for daily weaning assessment, anticoagulation targets, equipment monitoring, and frequency of neurological and limb monitoring should be adhered to with the goal of minimizing complication risk and allowing for early recognition if these complications occur.

ACKNOWLEDGMENTS

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Fig. 32.9 Femoral arterial and venous exposure (A) and cannulation (B).

Fig. 32.10 Landmark for subclavian vein access for subclavian vein cannulation.